

How Big Should My Cohort Be?

Sample-Size Estimation Across the Prediction-Model Lifecycle

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Binary-outcome clinical prediction models (e.g. pCR)

The question that keeps coming back

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The tempting shortcut

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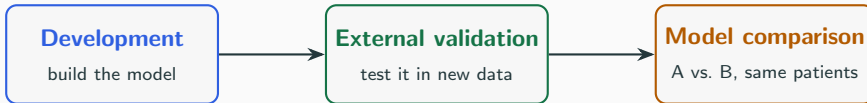
The tempting shortcut

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Why methodologists moved on (Riley, Snell, van Smeden, Collins, Jung)

Required N depends jointly on the **event fraction**, **number of predictors**, **expected performance** (R^2 , c) and the **target precision** — one-size-fits-all rules are often *too small* (imprecise) or *too big* (wasteful).

One study, three (or four) sample-size questions



Q1. Is the training set big enough to fit a *stable* model (little overfitting)?

Q2/Q3. Is the validation cohort big enough to estimate performance *precisely*?

Q4. Is the cohort big enough to *detect* that model B beats model A?

This is also the organising logic of TRIPOD+AI (item 10 asks for sample-size justification at development *and* validation).

Four methods, one timeline

Script	Stage	Method paper	Year
ss_3	Development	Riley et al., <i>BMJ</i> m441	2020
ss_2	Validation (sim.)	Snell et al., <i>J Clin Epidemiol</i>	2021
ss_1	Validation (closed)	Riley et al. (part 3), <i>BMJ</i>	2023
ss_4	Comparison	Jung, <i>Pharm Stat</i>	2024

Talk order = execution order

Develop → Validate (closed form + simulation as a cross-check) → Compare.

Development

ss_3 · Riley 2020

Development: four criteria (all reverse-lookup at your N)

Goal: minimise overfitting *before* you ever validate. Symbols: φ event fraction, P candidate parameters, R_{cs}^2 Cox–Snell R^2 , S shrinkage.

B1 — estimate φ precisely

$$\delta_{\text{margin}} = 1.96 \sqrt{\varphi(1 - \varphi)/N}$$

B2 — small prediction error (MAPE)

$$\ln \text{MAPE} = -0.508 - 0.544 \ln N + 0.259 \ln \varphi + 0.504 \ln P$$

B3 — shrinkage $S \geq 0.9$ (usually binding)

$$N = \frac{P}{(S - 1) \ln(1 - R_{cs}^2/S)}$$

B4 — optimism in $R_{\text{Nagelkerke}}^2 \leq 0.05$

$$\text{via } S_{B4} = \frac{R_{cs}^2}{R_{cs}^2 + \delta \max(R_{cs}^2)}$$

$$\max(R_{cs}^2) = 1 - \{\varphi^\varphi(1 - \varphi)^{1-\varphi}\}^2. \quad \text{Required } N = \max(N_{B1}, N_{B2}, N_{B3}, N_{B4}).$$

Development: what you feed it, what it tells you

Inputs (all scalars — no CSV needed)

- φ : anticipated pCR fraction
- P : candidate parameters
(fusion models: use a conservative proxy, not the neural-net weight count!)
- R_{CS}^2 : default $0.15 \times \max(R_{CS}^2)$

Worked example

($N = 717$, $\varphi = 0.30$, $P = 20$)

- B3 shrinkage $S \approx 0.80$ (target 0.90)
- Required $N \approx \mathbf{1599}$, binding = B3

$\Rightarrow$ development set *short* ≈ 880 patients

Takeaway

The shrinkage criterion (B3) almost always dominates — report it as your binding constraint.

External validation ss_2/ss_1 · Snell

2021 / Riley 2023

Validation: from “required N ” to “achieved precision”

Model is *already built*. Question: are the performance estimates in new data **precise enough**?

We report the achieved **95% CI width** for four measures, against community targets:

Measure	Target 95% CI width	Typically...
c-statistic (discrimination)	≤ 0.10	easy to meet
Calibration slope	≤ 0.30	the binding one
O/E ratio (calibration-in-large)	≤ 0.22	moderate
Standardised net benefit	context	threshold-specific

Two engines that should agree: `ss_1` closed-form (fast) and `ss_2` simulation (flexible).

ss_1: closed-form (the pmvalsampsize route)

c-statistic — Newcombe's formula (no iteration)

$$SE(C) = \sqrt{\frac{C(1-C) \left[1 + \left(\frac{N}{2} - 1\right) \frac{1-C}{2-C} + \left(\frac{N}{2} - 1\right) \frac{C}{1+C} \right]}{N^2 \varphi(1-\varphi)}}$$

O/E ratio

$$SE(\ln O/E) = \sqrt{\frac{1-\varphi}{N\varphi}}$$

Calibration slope

Fisher information of (α, λ) over the LP distribution (well-calibrated: $\alpha=0, \lambda=1$).

Inputs: N , φ , assumed c ; optionally a Beta shape for the predicted-risk histogram.

ss_2: simulation (Snell 2021) — when assumptions bite

1. Assume the linear predictor $LP_i \sim \mathcal{N}(\mu, \sigma^2)$ in the validation population.
2. Impose a *true* calibration model: $\text{logit}(p_i) = \gamma + S \cdot LP_i$.
 - $\gamma = 0, S = 1$: well calibrated
 - $S < 1$ (e.g. 0.9): predictions too extreme — *stress-test* miscalibration
3. Simulate B datasets, refit, average the 95% CI widths.

Why keep both

ss_1 is instant and matches software; ss_2 lets you inject miscalibration and use your *real* LP distribution. Agreement between them is a good sanity check.

Worked example ($N = 334$): c -width ≈ 0.11 (OK) but slope width $\approx 0.5-0.6$ (wide) $\Rightarrow$ slope binds.

Bridge: turning a reported c -statistic into R_{cs}^2

Development (ss_3) wants R_{cs}^2 ; you often only *have* a c -statistic from a pilot/prior model.

$$\delta = \sqrt{2} \Phi^{-1}(C) \quad \implies \quad R_{cs}^2 \approx 1 - \exp\left(-\frac{\delta^2}{2}\right) \text{ (approx.)}$$

Caveat

This bridge assumes normal linear predictors — an *approximation*. Use it as a conservative starting point; prefer `pmsampsize`'s built-in conversion for manuscript numbers.

Model comparison

ss_4 • Jung 2024

Comparing two AUCs on the same patients (paired)

Question shifts from “how good is model B?” to “is B *significantly* better than A?” — the sample-size version of **DeLong’s test**.

Standardised effect $\leftrightarrow$ AUC

$$X_k \sim \mathcal{N}(\delta_k, 1) \text{ (cases)}, Y_k \sim \mathcal{N}(0, 1) \text{ (controls)}, \quad \theta_k = \Phi(\delta_k/\sqrt{2})$$

Required N

$$N = \frac{v(z_{1-\alpha/2} + z_{1-\beta})^2}{(\theta_1 - \theta_2)^2}$$

$$v = \frac{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}{\gamma(1-\gamma)}, \text{ computed by numerical integration.}$$

Reverse: power at your N

$$1 - \beta = \Phi\left(\frac{\sqrt{N}|\Delta|}{\sqrt{v}} - z_{1-\alpha/2}\right)$$

Key extra input: ρ , the correlation between the two models’ scores (paired design). Worked: $\theta=0.75$ vs 0.82 ,

$\rho=0.5 \Rightarrow N \approx 422$; power at $N=334$ only ≈ 0.70 .

Putting it together

What each script needs — at a glance

Script	Core question	Key inputs	Key output
ss_3	training set big enough?	φ, P, R_{cs}^2, N	required N ; binding B1–B4
ss_2	validation precise enough?	$\mu, \sigma, \gamma, S, N$	avg 95% CI widths (sim.)
ss_1	validation precise enough?	φ, c, N	achieved 95% CI widths
ss_4	B beats A detectable?	$\theta_1, \theta_2, \rho, \gamma, N$	required N ; power; MDD

All four support a reverse lookup

Plug in your actual cohort size → read off achieved precision / power. No guessing N up front.

Three drop-in paragraphs (development / validation / comparison). Skeleton:

Validation example

“Following Riley et al. (2023) and, as a sensitivity check, the simulation approach of Snell et al. (2021), assuming $c = \dots$ and event proportion = $\dots$, the N required for 95% CI widths of ≤ 0.10 (c), ≤ 0.30 (slope) and ≤ 0.22 (O/E) was $\dots$ ($\dots$ events), driven by the calibration slope. Our cohort ($N = \dots$) achieved a slope CI width of $\approx \dots$, indicating [precise / limited] precision.”

The repository ships all three templates ready to fill with your numbers.

Software: a package + a zero-install web demo

sampsizeval (Python)

- 4 calculators, one API
- CLI: `sampsizeval dev/val/compare`
- CSV $\rightarrow$ parameters (φ, c, ρ)
- pytest test suite

Browser demo (Pyodide)

- Runs 100% client-side — data never leaves the machine
- Manual inputs *or* drag-in a CSV
- Hosted free on GitHub Pages

Try it

<https://csyfjiang.github.io/Sample-Size-Estimation-Tutorial/>

Takeaways

1. Sample size is a **lifecycle** question: develop $\rightarrow$ validate $\rightarrow$ compare.
2. Drop the rules of thumb — required N depends on φ , P , expected performance and *target precision*.
3. **Development**: shrinkage (B3) is the usual binding constraint.
4. **Validation**: the *calibration slope* CI width usually binds, not the c -statistic.
5. **Comparison**: correlation ρ between models drives the required N .
6. Always report the **reverse lookup** for your actual cohort (TRIPOD+AI item 10).

- Riley RD, Ensor J, Snell KIE, et al. Calculating the sample size required for developing a clinical prediction model. *BMJ* 2020;368:m441.
- Snell KIE, Archer L, Ensor J, et al. External validation of clinical prediction models: simulation-based sample size calculations were more reliable than rules-of-thumb. *J Clin Epidemiol* 2021;135:79–89.
- Riley RD, Snell KIE, Archer L, et al. Evaluation of clinical prediction models (part 3): calculating the sample size required for an external validation study. *BMJ* 2023;383:e074821.
- Jung SH. Sample size calculation for comparing two ROC curves. *Pharmaceutical Statistics* 2024;23(4):557–569.
- van Smeden M, et al. Sample size for binary logistic prediction models. *Stat Methods Med Res* 2019.

Thank you

`https://github.com/csyfjiang/Sample-Size-Estimation-Tutorial`